Problem statement

Send feedback

A network having a bandwidth of 12Mbps can pass 10,000 frames per minute. Each frame is of size 12,000 bits. Calculate the throughput of the network? Specify the answer in Mbps and enter the integer part of the answer.

Since 10000 frames per minute can pass and each frame is of 12,000 bits, the number of bits passing per second is: (10000*12000)/60 =2,000,000 bps = 2Mbps So, the throughput is 2Mbps which is almost 1/5th of the bandwidth of the network.

Choose the replication strategy (out of active and passive replication), which should be preferred for the following scenarios:
B. If Netflix wants to know how many people viewed a particular movie.

In this kind of scenario active replication should be preferred. Since this computation cannot be handled by one single master mode, all the active nodes need to do the write operation and then later sync within themselves to know the exact number.

Choose the correct load balancing algorithms in which client requests are distributed to the server with the minimum number of active connections for different server characteristics.

Weighted Least Connection builds on the Least Connection load balancing algorithm to account for differing application server characteristics. The administrator assigns a weight to each application server based on criteria of their choosing to demonstrate the application server traffic-handling capability.

In the context of system design, which of the following caching strategies is used in most relational database storage engines with write-heavy workloads?

Write back is a storage method in which data is written into the cache every time a change occurs, but is written into the corresponding location in the main memory only at specified intervals or under certain conditions. This is sometimes called write-behind as well. Write back caches improve the write performance and are good for write-heavy workloads.

Suppose there is a table(Students) with the SELECT Query given below. Which column can be considered for indexing in reference to the query?

```
SELECT
student_id,
class
FROM Students
WHERE first_name='Vishwa'
```

student_id and class are referenced in the SELECT portion and MySQL will not use them to limit the chosen rows. Therefore, indexing them will make no difference. first_name is referenced in the WHERE clause and it will be used to search the particular record. So, first_name should be indexed for query optimization and faster searching.

Two servers are communicating with each other in real-time. Server A sends MX types of messages and server B sends MY types of messages. The processing time of MX is 10ms and MY is 4ms. The network round trip time is 2ms. The sending of MX types and MY types are executed serially. A and B are single-threaded. What will be the best case overall latency of sending and acknowledging 10 MX and 10 MY messages from A to B?

Latency = Processing time + Network latency Here, Network latency = 2ms. Processing time of MX = 10ms. Processing time of MY = 4ms. Total time = (count of MX)(latency of MX) + (count of MY)(latency of MY) = 10*(10+2) + 10*(4+2) = 120 + 60 = 180ms

Suppose both A and B are multi-threaded. What will be the best case latency then?

Asynchronous communication is possible here since the asynchronous model allows multiple calls to be executed at the same time. The latency would therefore be the maximum of the latency of MX and latency of MY because first calls can execute on separate threads while second calls are pending execution. Latency = 10*(10+2) = 120ms

Ram wants to build an online shopping application that supports XML and JSON using the REST API. Which HTTP request method should be used for:
Request to /user/ returns a list of registered users on a system
Specify the answer in capital letters

GET

(successful requests → HTTP status code: 200)

We want to design an authentication system which is an online shopping application. How would the basic authentication work for the system?

Basic authentication is the simplest type of authentication. It would consist of the following steps: 1. Fetch the user credentials including username and password from the user. 2. Encode it using base64 encoding schemes. 3. Send it along with the HTTP request by attaching it with the authorization header.

Challenges :

Since the credentials are encoded using base64 encoding scheme, it is easy to decode it and therefore poses huge security issues. It is also not safe to send confidential information like password over the HTTP request.

Token authentication is the stateless type of authentication. It would consist of the following steps: 1. Fetch the user credentials including username and password from the user. 2. Using form parameters, it is sent to the server. 3. Authorization takes place on the server using the credentials received from step2. 4. If the validation is successful, the server creates a signed token(can be used by the user to verify identity without entering credentials) JWT(JSON Web Token) which is sent as a response. 5. The Token is used for all requests in future then.

Challenges

If once validated, the access of the user cannot be revoked. It is important to safely handle the token and can be prone to security issues.

OAuth is the extension of the token-based type of authentication. It would consist of the following steps: 1. The authentication request is sent by the user. 2. The server validates the user by checking for an existing account and returning back with an authorization grant. 3. The application requesting authentication then uses the grant for accessing information. 4. On authorization, a token is generated for the access. 5. The resource can then be accessed using this token.